



Life on Mars indicated

NASA's Curiosity rover has detected copious amounts of methane gas on Mars which may indicate alien life. <https://digit.in/july19-16>



Car-sized giant squid found

A giant squid has been caught on camera in US waters living 2500 feet below the surface, in the 'midnight zone'. <https://digit.in/july19-17>

THE FERMİ PARADOX

Where are they?

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Although full of life, the forest at night is eerily silent

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strobiology research papers paint a picture of a universe that is excessively conducive to life. Soon after the

big bang, the ambient conditions were such that all young rocky planets perpetually basked in the habitable zone. Liquid water could exist on the surface of any rocky planet, irrespective of its distance from the host star. Analysis of space dust has shown that there is a spike in the volume of dust at a particular size. Dormant bacteria is exactly that size. The implication is that the universe may be seeded with dormant bacteria, waiting to find a suitable planet to live in. A thought exercise in mapping the increase in complexity of life to Moore's law, showed a chilling match. Life on Earth increases in complexity more or less along a straight line, only that line extends to before the Earth was formed. Researchers found building blocks of life in nebulae and asteroids.

Objects such as 'Oumuamua, which bounce between the stars in a galaxy, can spread life from one star to another. Eventually, such objects can introduce life to an entire galaxy.

The Statistical Drake Equation (SDE), a mathematically stronger variant of the Drake equation, allows researchers to estimate the number of advanced civilisations emerging from Earth-like planets in orbit around Sun-like stars. These are civilisations of intelligent beings capable of interstellar communication. There are estimated to be 4,590 such civilisations in the Milky Way.

Statistics indicate that every star has two rocky planets in the habitable zone, on average. Earth may well be an outlier case when it comes to suitability for life. There can be super-habitable planets, where the conditions are even more conducive to life. On a galactic or universal scale, life may be most common in ice worlds around gas giants, such as Europa and Ence-

ladus. There are 100 to 1000 times as many ice worlds in our galaxy, as there are rocky planets in hab zone orbits. The atmosphere in circumbinary planets (worlds with two stars in the sky) are stable for longer than planets such as Earth. Circumbinary planets may thus be the best niches for life in the Universe. All of this, is for oxygen-breathing carbon-based life forms that depend on liquid water. There might be other forms of life as well.

These can be silicon or sulfur-based life forms. Instead of water, they may use ammonia, hydrocarbons such as methane, silicon dioxide or hydrogen fluoride. Liquid nitrogen can be a solvent at very low temperatures, while sodium chloride can perform the same function at very high temperatures. The conditions on some worlds are such that they can support two forms of life. The hydrocarbon lakes on the Saturnian moon, Titan, can support a methane-based life form in its hydrocarbon lakes. These hypothetical